

# Project Goal

## End-to-End Retail Data Engineering Pipeline

Tech Stack:

- Python
- Pandas
- PostgreSQL
- Docker
- Apache Airflow
- Power BI
- GitHub



## WEEK 1 — Foundation & Setup

### Common Goal

Understand the project architecture and set up the development environment.

#### Team Member 1

- Install Python
- Learn Pandas basics
- Read retail dataset
- Explore columns and data types
- Create GitHub repository
- Learn ETL concepts

Deliverable:

- Read CSV using Python
- Basic data profiling report

#### Team Member 2

- Install PostgreSQL
- Install pgAdmin
- Learn basic SQL
- Create first database
- Understand Star Schema concepts

Deliverable:

- retail\_dw database
- Basic SQL queries

# Joint Deliverable

- Project Architecture Diagram
- Technology Understanding
- GitHub Repository Setup

## WEEK 2 — Data Cleaning & Warehouse Design

### Common Goal

Convert raw data into structured warehouse design.

### Team Member 1

- Handle missing values
- Remove duplicates
- Standardize data types
- Create ETL script

Deliverable:

```
clean_data.py
```

### Team Member 2

Design warehouse tables:

```
Fact_Sales
Dim_Product
Dim_Location
Dim_Segment
Dim_Shipping
```

Create tables in PostgreSQL.

Deliverable:

```
warehouse_schema.sql
```

# Joint Deliverable

Star Schema Completed



## WEEK 3 — ETL Pipeline Development

### Common Goal

Load cleaned data into PostgreSQL.

### Team Member 1

Develop:

```
extract.py
transform.py
load.py
```

Connect Python to PostgreSQL.

Deliverable:

```
Working ETL Pipeline
```

### Team Member 2

- Create primary keys
- Create foreign keys
- Validate loaded data
- Write warehouse queries

Deliverable:

```
Fact and Dimension tables populated
```

### Joint Deliverable

```
CSV → PostgreSQL Pipeline Working
```



## WEEK 4 — Docker & Airflow

### Common Goal

Automate and containerize the project.

### Team Member 1

Learn Airflow:

- DAG
- Tasks
- Operators

- Scheduling

Create:

```
retail_etl_dag.py
```

Deliverable:

```
Airflow ETL Workflow
```

## Team Member 2

Learn Docker:

- Images
- Containers
- Docker Compose

Containerize:

```
PostgreSQL
Airflow
```

Deliverable:

```
docker-compose.yml
```

## Joint Deliverable

```
Docker + Airflow Integration
```

# WEEK 5 — Dashboard, Testing & Documentation

## Common Goal

Complete project and prepare presentation.

## Team Member 1

- Test ETL workflow
- Fix pipeline issues
- Complete documentation
- Prepare architecture diagrams

Deliverable:

```
Technical Documentation
```

## Team Member 2

Build Power BI Dashboard:

- Sales Analysis
- Profit Analysis
- Category Performance
- Region Performance

Deliverable:

Final Dashboard

## Joint Deliverable

Final Presentation  
GitHub Repository  
Project Report  
Demo Video (Optional)

## 🏁 Final Deliverables

By the end of Week 5:

- ✓ Python ETL Pipeline
- ✓ PostgreSQL Data Warehouse
- ✓ Star Schema
- ✓ Dockerized Environment
- ✓ Apache Airflow DAG
- ✓ Power BI Dashboard
- ✓ GitHub Repository
- ✓ Project Documentation